

Lists and List Slicing

Learning Objectives

- Create and access Python lists.
- Use positive and negative indexing.
- Understand list slicing using start and stop positions.
- Use the step value in slicing.
- Reverse lists using slicing.
- Modify, insert, delete, and replace list elements using slices.

Important Slicing Rule

Basic syntax: list[start : stop]

The start index is included, but the stop index is excluded.

Extended syntax: list[start : stop : step]

Basic List Experiments

1. Create a list

```
fruits = ["apple", "banana", "mango", "orange"]  
print(fruits)
```

2. Find length of a list

```
numbers = [10, 20, 30, 40, 50]  
print(len(numbers))
```

3. Access first element

```
numbers = [10, 20, 30, 40, 50]  
print(numbers[0])
```

4. Access last element

```
numbers = [10, 20, 30, 40, 50]  
print(numbers[-1])
```

5. Access second element

```
numbers = [10, 20, 30, 40, 50]  
print(numbers[1])
```

6. Access third element from the end

```
numbers = [10, 20, 30, 40, 50]  
print(numbers[-3])
```

7. Change an element

```
numbers = [10, 20, 30, 40, 50]  
numbers[2] = 100  
print(numbers)
```

8. Add an element using append()

```
numbers = [10, 20, 30]  
numbers.append(40)  
print(numbers)
```

9. Insert an element

```
numbers = [10, 20, 30]
numbers.insert(1, 15)
print(numbers)
```

10. Remove an element

```
numbers = [10, 20, 30, 40]
numbers.remove(30)
print(numbers)
```

Basic Slicing

11. Slice first three elements

```
numbers = [10, 20, 30, 40, 50]
print(numbers[0:3])
```

12. Slice elements 2 to 4

```
numbers = [10, 20, 30, 40, 50]
print(numbers[1:4])
```

13. Slice from beginning

```
numbers = [10, 20, 30, 40, 50]
print(numbers[:3])
```

14. Slice to the end

```
numbers = [10, 20, 30, 40, 50]
print(numbers[2:])
```

15. Copy the entire list using slicing

```
numbers = [10, 20, 30, 40, 50]
copy = numbers[:]
print(copy)
```

16. Slice using negative index

```
numbers = [10, 20, 30, 40, 50]
print(numbers[-4:-1])
```

17. Last three elements

```
numbers = [10, 20, 30, 40, 50]
print(numbers[-3:])
```

18. First four elements

```
numbers = [10, 20, 30, 40, 50]
print(numbers[:4])
```

19. Middle elements

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[2:5])
```

20. Select elements from index 1 to 3

```
numbers = [10, 20, 30, 40, 50]
print(numbers[1:4])
```

Slicing with Step

21. Take every second element

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[::2])
```

22. Take every third element

```
numbers = [10, 20, 30, 40, 50, 60, 70]
print(numbers[::3])
```

23. Start from index 1, take every second element

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[1::2])
```

24. Slice with step 2

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[0:5:2])
```

25. Slice with step 3

```
numbers = [10, 20, 30, 40, 50, 60, 70]
print(numbers[1:7:3])
```

26. Reverse a list

```
numbers = [10, 20, 30, 40, 50]
print(numbers[::-1])
```

27. Reverse using negative step

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[5:1:-1])
```

28. Print alternate elements in reverse

```
numbers = [10, 20, 30, 40, 50, 60]
print(numbers[::-2])
```

29. Reverse first four elements

```
numbers = [10, 20, 30, 40, 50]
print(numbers[3::-1])
```

30. Reverse last three elements

```
numbers = [10, 20, 30, 40, 50]
print(numbers[:1:-1])
```

String Lists and Slicing

31. Slice a list of names

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha"]
print(names[1:4])
```

32. First two names

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha"]
print(names[:2])
```

33. Last two names

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha"]
print(names[-2:])
```

34. Alternate names

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha", "Farhan"]
print(names[::2])
```

35. Reverse names

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha"]  
print(names[::-1])
```

36. Select names from index 1 to 3

```
names = ["Arun", "Bala", "Charan", "Deepak", "Esha"]  
print(names[1:4])
```

37. Slice subjects

```
subjects = ["Python", "DBMS", "Cloud", "AI", "Gaming", "Data Science"]  
print(subjects[2:5])
```

38. Alternate subjects

```
subjects = ["Python", "DBMS", "Cloud", "AI", "Gaming", "Data Science"]  
print(subjects[::2])
```

39. Reverse subjects

```
subjects = ["Python", "DBMS", "Cloud", "AI", "Gaming", "Data Science"]  
print(subjects[::-1])
```

40. Select last four subjects

```
subjects = ["Python", "DBMS", "Cloud", "AI", "Gaming", "Data Science"]  
print(subjects[-4:])
```

Slicing for Modification

41. Replace two elements

```
numbers = [10, 20, 30, 40, 50]  
numbers[1:3] = [200, 300]  
print(numbers)
```

42. Replace three elements with one element

```
numbers = [10, 20, 30, 40, 50]  
numbers[1:4] = [100]  
print(numbers)
```

43. Replace one element with multiple elements

```
numbers = [10, 20, 30, 40]  
numbers[1:2] = [200, 300, 400]  
print(numbers)
```

44. Delete elements using slicing

```
numbers = [10, 20, 30, 40, 50]  
numbers[1:4] = []  
print(numbers)
```

45. Insert elements using slicing

```
numbers = [10, 20, 40, 50]  
numbers[2:2] = [25, 30, 35]  
print(numbers)
```

46. Replace alternate elements

```
numbers = [10, 20, 30, 40, 50, 60]  
numbers[::2] = [100, 300, 500]  
print(numbers)
```

47. Delete alternate elements

```
numbers = [10, 20, 30, 40, 50, 60]
numbers[::2] = []
print(numbers)
```

48. Make a copy and modify it

```
numbers = [10, 20, 30, 40, 50]
new_numbers = numbers[:]
new_numbers[0] = 999
print("Original:", numbers)
print("Copy:", new_numbers)
```

49. Extract even-position elements

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80]
print(numbers[1::2])
```

50. Extract odd-position elements

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80]
print(numbers[::2])
```